Fixture record fx-0013

Title: Discrete-event simulation of a rural community's mass-casualty triage pathway

Abstract:
A discrete-event simulation of the mass-casualty triage pathway at a single rural community hospital is used to test staffing configurations against a simulated multi-vehicle accident and a simulated tornado strike.

Methods
Study design: Simulation study (discrete-event simulation)
Country: United States
Population: Staff and patients at a single rural community hospital
Concept: Discrete-event simulation of mass-casualty triage pathway staffing
Context: Rural mass-casualty and tornado-strike disaster response

Results
- Adding one additional triage nurse reduced simulated time-to-triage by 31% under the tornado-strike scenario.
- The multi-vehicle-accident scenario did not saturate triage capacity at current staffing levels.

Limitations
The authors state that the model did not account for staff absenteeism during the simulated disaster itself.

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Funder role: (not stated)

Synthetic test data — pipeline conformance fixture, not scientific evidence.